

Novex Pharma · EQMS Walkthrough

Persona-driven UI walkthrough + AI agent outputs

Tenant: Novex Pharma · Backend: local (demo data seeded) · Frontend: Vercel live · Captures: 19

Personas: 6

UI screenshots: 10

AI agent outputs: 9

Generated: 2026-04-23

Contents

- **Kenji** · QA Specialist · 7 screenshots
- **Priya** · Audit Program Manager · 4 screenshots
- **Maria** · Lead Auditor · 3 screenshots
- **James** · Head of QA · 2 screenshots
- **Marcus** · Regulatory Affairs · 1 screenshots
- **Elena** · VP of Quality · 2 screenshots

Kenji

QA Specialist

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#01

Dashboard after login

UI screenshot

QA Specialist lands on the Novex dashboard. Sidebar shows EQMS modules.

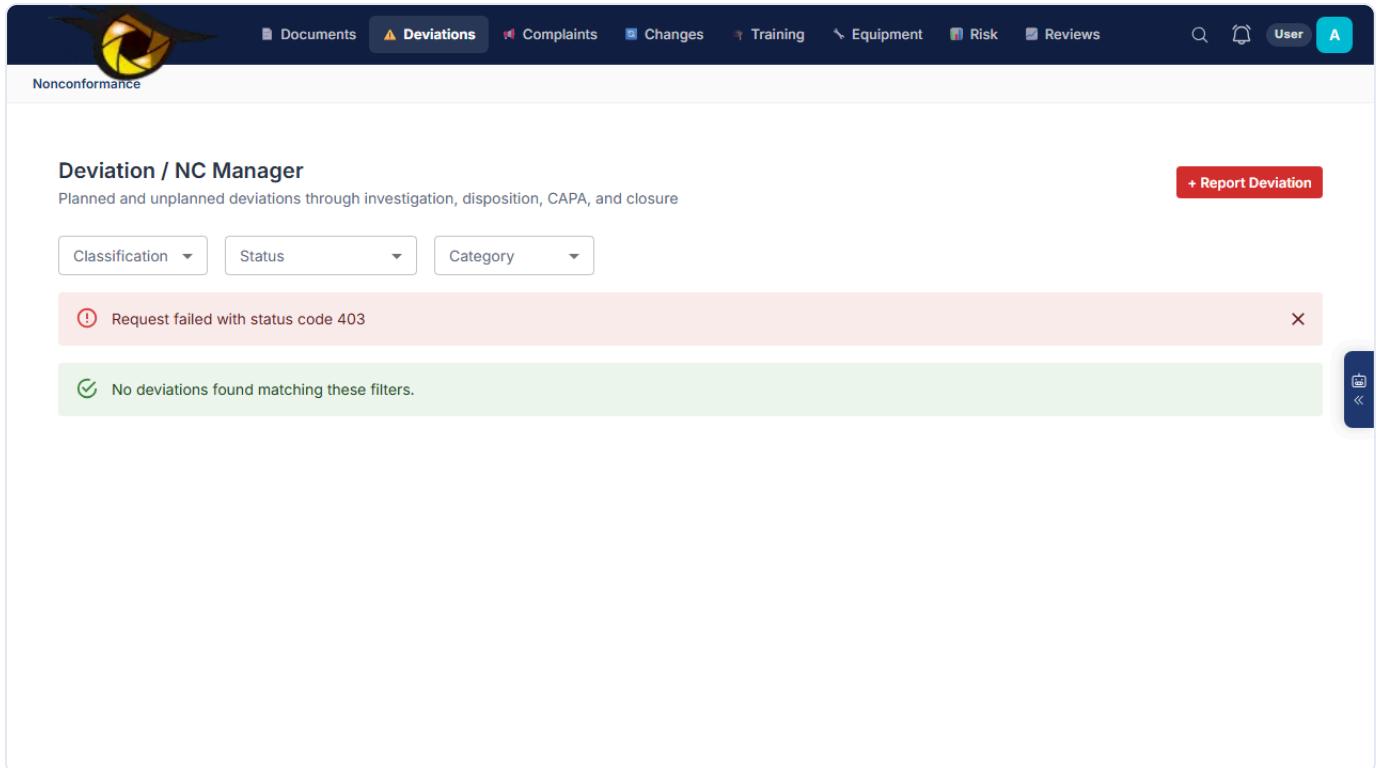


#02

Deviation detail (attempt)

UI screenshot

Drilled into a deviation detail view from the non-conformance register.



The screenshot displays the 'Deviation / NC Manager' interface within a 'Nonconformance' section. The top navigation bar includes links for Documents, Deviations (active), Complaints, Changes, Training, Equipment, Risk, and Reviews, along with search, notification, and user profile icons. The main content area features a title 'Deviation / NC Manager' and a subtitle 'Planned and unplanned deviations through investigation, disposition, CAPA, and closure'. A red '+ Report Deviation' button is located in the top right. Below the title, there are three filter dropdowns: 'Classification', 'Status', and 'Category'. A red error message bar states 'Request failed with status code 403'. Below this, a green success message bar states 'No deviations found matching these filters.' A vertical sidebar on the right contains a list icon and a back arrow.

Nonconformance

Deviation / NC Manager

Planned and unplanned deviations through investigation, disposition, CAPA, and closure

+ Report Deviation

Classification Status Category

Request failed with status code 403

No deviations found matching these filters.

CAPA module showing seeded CAPA-DEMO-001 (major, DRAFT) and CAPA-DEMO-002 (minor, APPROVED).



Documents

Search

Notifications

UserA

Buyer > CAPAS

CAPA Management Workspace

OPEN0

OVERDUE0

RECURRING0

NEW CANDIDATES0

Audit ID

Generate

Manual CAPA title

Create

Search candidates

Candidate StatusAll

Q Fuzzy search ⓘ

Search CAPAs

CAPA StatusAll

Refresh

Q Fuzzy search ⓘ

CAPA Detail

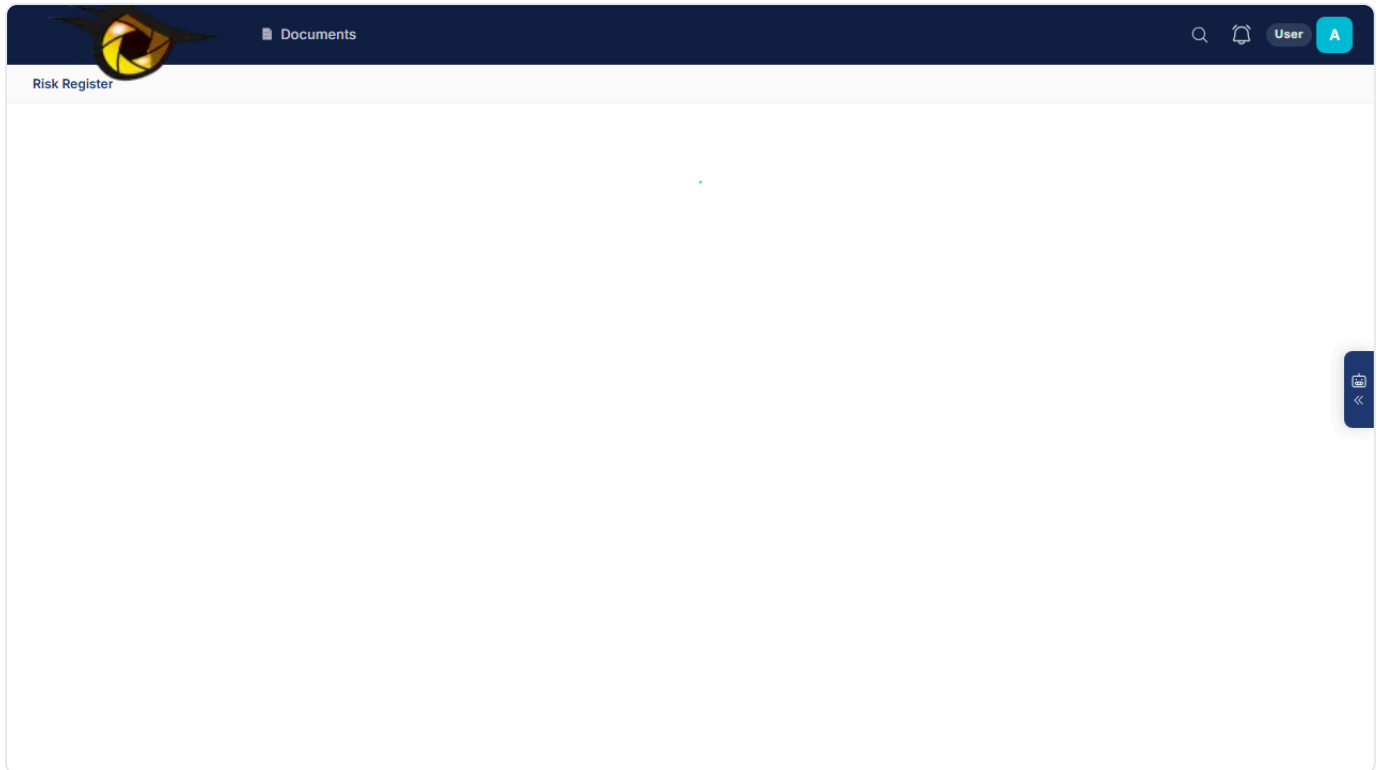
ⓘ Select a CAPA to review details.

#04

Risk Register

UI screenshot

FMEA-driven risk register. QA Specialist raises and mitigates risks here.



#101

AI output · Deviation 5-why scaffolder

AI output

AI scaffolds the 5-why, categorises the investigation (process/equipment/material), and suggests specific follow-ups to raise on the shop floor. Grounded on the deviation narrative. Confidence-gated.

AI output · Deviation 5-why scaffolder

Kenji (QA Specialist) lets the AI scaffold the investigation

Called by: Kenji

endpoint: POST /api/ai/deviation/scaffold-five-why

confidence: —

model: —

latency: 0ms

Agent fell back: **low confidence**. I could not verify this with enough confidence from the controlled documents and evidence available. Please provide additional context, or draft this section manually and I will help review it.

AI returns a fully structured CAPA: 5-why with citations, corrective + preventive actions (each tagged with owner role + due days), effectiveness check, regulatory clauses. Every claim cites a source.

AI output · CAPA RCA drafter

Full RCA grounded on SOP + prior CAPA + FDA corpus

Called by: Kenji

endpoint: POST /api/ai/capa/draft-rca

confidence: 0.70

severity: major

model: gemini-2.5-flash-lite

citations: 3

EXECUTIVE NARRATIVE

Batch NVX-2026-B014 failed its dissolution specification, with results of 76% and 74% compared to the NLT 80% requirement. This follows two similar occurrences on the same manufacturing line (Line 2) in the previous quarter. A prior investigation (PRIOR-CAPA-042) had previously linked dissolution OOS results on Line 2 to blending-time drift. However, the root cause for the blending-time drift itself has not been identified or addressed, indicating a systemic issue with process control for blending time on this line.

5-WHY CHAIN

Why 1: Why did Batch NVX-2026-B014 fail dissolution specification?

→ The batch failed dissolution testing, yielding 76% (retest 74%) against the acceptance criterion of NLT 80%. [DEVIATION NARRATIVE](#)

Why 2: Why did the batch exhibit low dissolution results?

→ Similar low dissolution results (2 instances) were observed last quarter on the same manufacturing line (Line 2). [DEVIATION NARRATIVE](#)

Why 3: Why were there similar low dissolution results on Line 2 last quarter?

→ A prior CAPA (PRIOR-CAPA-042) identified that dissolution Out-of-Specification (OOS) results on Line 2 were traced to blending-time drift. [SOURCE_2:findings](#)

Why 4: Why did blending-time drift occur on Line 2?

→ The provided sources do not contain information explaining the cause of the blending-time drift itself.

CORRECTIVE ACTIONS

corrective Review batch records for NVX-2026-B014 to confirm all processing steps were followed as per approved batch record.

owner: QA Specialist · due in 14d · ref: [DEVIATION NARRATIVE](#)

corrective Perform a targeted investigation into the specific blending parameters and duration for Batch NVX-2026-B014.

owner: Manufacturing Supervisor · due in 14d · ref: [SOURCE_2:findings](#)

PREVENTIVE ACTIONS

preventive Implement enhanced monitoring and control measures for blending time on Line 2, including calibration verification and drift detection alarms.

owner: Head of Manufacturing · due in 30d · ref: [SOURCE_2:findings](#)

preventive Review and update SOPs related to blender operation and time control on Line 2 to prevent recurrence of blending-time drift.

owner: Head of Manufacturing · due in 30d · ref: [SOURCE_2:findings](#)

EFFECTIVENESS CHECK

method	Monitor dissolution results for all subsequent batches manufactured on Line 2 for a period of 90 days. Review blending time logs for Line 2 for any recorded deviations or drifts.
success criteria	No dissolution failures attributed to blending time drift and no recorded blending time drifts on Line 2 for 90 days.
review after	90 days

REGULATORY CLAUSES CITED

[21 CFR 211.100](#) [21 CFR 211.192](#) [ICH Q10 §3.2.4](#) [ICH Q9](#)

#103

AI output · Predictive CAPA effectiveness badge

AI output

Shows P(on-time) and P(effective) with top factors. Used as decision-support — appears inline next to the CAPA owner field so Kenji can adjust slack before approving.

AI output · Predictive CAPA effectiveness badge

Inline badge on every CAPA · calibrated heuristic

Called by: Kenji

endpoint: POST /api/ai/predict/capa-outcome

P(on-time): 76%

P(effective): 72%

model: capa.heuristic@1.0.0

confidence: 0.95

OUTCOME PREDICTION**76%**

on-time closure

72%

effective at close

TOP FACTORS (WHAT DROVE THE PREDICTION)**slack_days** · positive · contribution 0.47**owner_prior_closure_rate** · positive · contribution 0.19**severity** · negative · contribution -0.08**MODEL**

Heuristic predictor v1.0 (swappable for LightGBM). Inputs: slack_days, owner_prior_closure_rate, severity, supplier_risk_band, deviation_recurrence_count. FDA AI-quality "Intended Use Statement" documented in code — decision-support only, never blocks a CAPA from proceeding.

Priya

Audit Program Manager

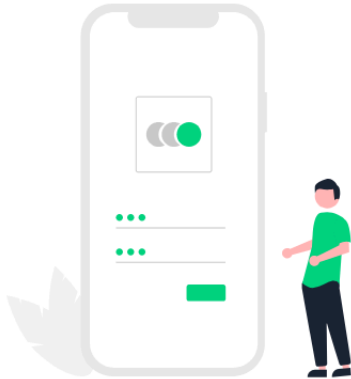
audit.program@novex-pharma.demo

#06

All internal + external audits

UI screenshot

Full audit register across the tenant. Each row links to its lifecycle detail.





Sign In to

Enter your email address & password to get started.



[Forgot Password?](#)

Sign In

OR

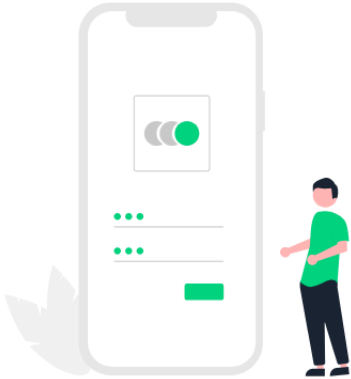
New to Hawkeye? [Create an Account](#)

#07

Suppliers register

UI screenshot

Registered suppliers. From here Priya can open any supplier to see the AI Supplier Intel dossier.





Sign In to

Enter your email address & password to get started.



[Forgot Password?](#)

Sign in

OR

New to Hawkeye? [Create an Account](#)

#104

AI output · Supplier Intel (public + tenant)

AI output

Searched 'Sun Pharmaceutical Industries'. Tenant card (blue) is empty — not in the Novex registry. Public card (orange) is rich — openFDA returned registered drugs + recalls, FDA warning letter search returned letters mentioning the firm. Provenance unambiguous.

AI output · Supplier Intel (public + tenant)

Live openFDA + FDA warning letter scrape on a real pharma firm

Called by: Priya

endpoint: POST /api/ai/audit-agents/supplier-intel

verdict: public_only

tenant matches: 0

public sources: openFDA, fdaWarningLetter, pharmaCompass

VERDICT · PROVENANCE NOTE

public_only Not in your supplier registry. Public data found — this entity is NOT one of your qualified suppliers.

TENANT (REGISTERED) DATA

Not in your supplier registry.

PUBLIC REGULATORY SIGNALS

0

FDA drug regs

1

recalls

6

warning letters

RECENT RECALLS

· **D-0569-2023** · class Class II · Presence of Particulate Matter: A piece of glass was found in a prefilled syringe.

WARNING LETTERS MENTIONING THIS FIRM

· About Warning and Close-Out Letters

· About Warning and Close-Out Letters

· Sun Pharmaceutical Industries Limited

AI pulls past findings + openFDA recalls + warning letters + baseline template, risk-weights each section (high/medium/low), flags FDA-visible risks (data integrity, OOS investigation, aseptic). Every question cites its source.

AI output · Audit Prep Agent (risk-weighted questionnaire)

Priya gets a tailored questionnaire in 10 seconds

Called by: Priya

endpoint: POST /api/ai/audit-agents/prepare-questionnaire

sections: 6

high-risk signals: 2

confidence: 0.90

PLAN SUMMARY

This audit plan prioritizes areas related to particulate matter contamination and quality control due to a recent FDA recall for glass particulate in prefilled syringes. Several warning letters also indicate a need for heightened scrutiny on data integrity and overall quality management systems.

HIGH-RISK SIGNALS FLAGGED FROM PUBLIC DATA

- FDA recall for particulate matter contamination (SOURCE_2)
- Multiple FDA warning letters (SOURCE_3)

RISK-WEIGHTED SECTIONS

high Quality Management Systems

Multiple FDA warning letters (SOURCE_3) indicate systemic issues within the quality management system, necessitating a high-priority review.

7 questions

high Manufacturing and Process Controls

The FDA recall (SOURCE_2) directly relates to particulate matter contamination during manufacturing, requiring a high-priority review of these controls.

6 questions

high Laboratory Controls

Warning letters often cite data integrity issues in laboratories, and particulate matter testing is critical for product quality.

6 questions

medium Documentation and Records

While not directly highlighted by the specific recall, comprehensive documentation is a cornerstone of GMP and often implicated in warning letters.

5 questions

medium Personnel

Adequate training and qualification of personnel are fundamental to GMP compliance and are often a contributing factor in quality issues.

3 questions

medium Facilities and Equipment

While the recall points to a specific issue, general facility and equipment controls are important for preventing future occurrences.

3 questions

Maria

Lead Auditor

audit.lead@novex-pharma.demo

#08 Auditor dashboard UI screenshot

Lead Auditor home with assigned audits and open findings count.



Audits

CAPAs

Documents

Search

Notifications

Auditor

M

Auditor > Dashboard

Auditor Dashboard

Assigned Audits

0

0 overdue

Report Drafts

0

0 upcoming

Issues (my review)

0

0 overdue

CAPAs (review)

0

0 overdue

My Work Queue

Refresh

Item	Status	Due / Target	Updated	Action
No items				

Assigned Audits

Detailed assigned audits list will render here. Use Audits tab for now.

Issue & CAPA Inbox

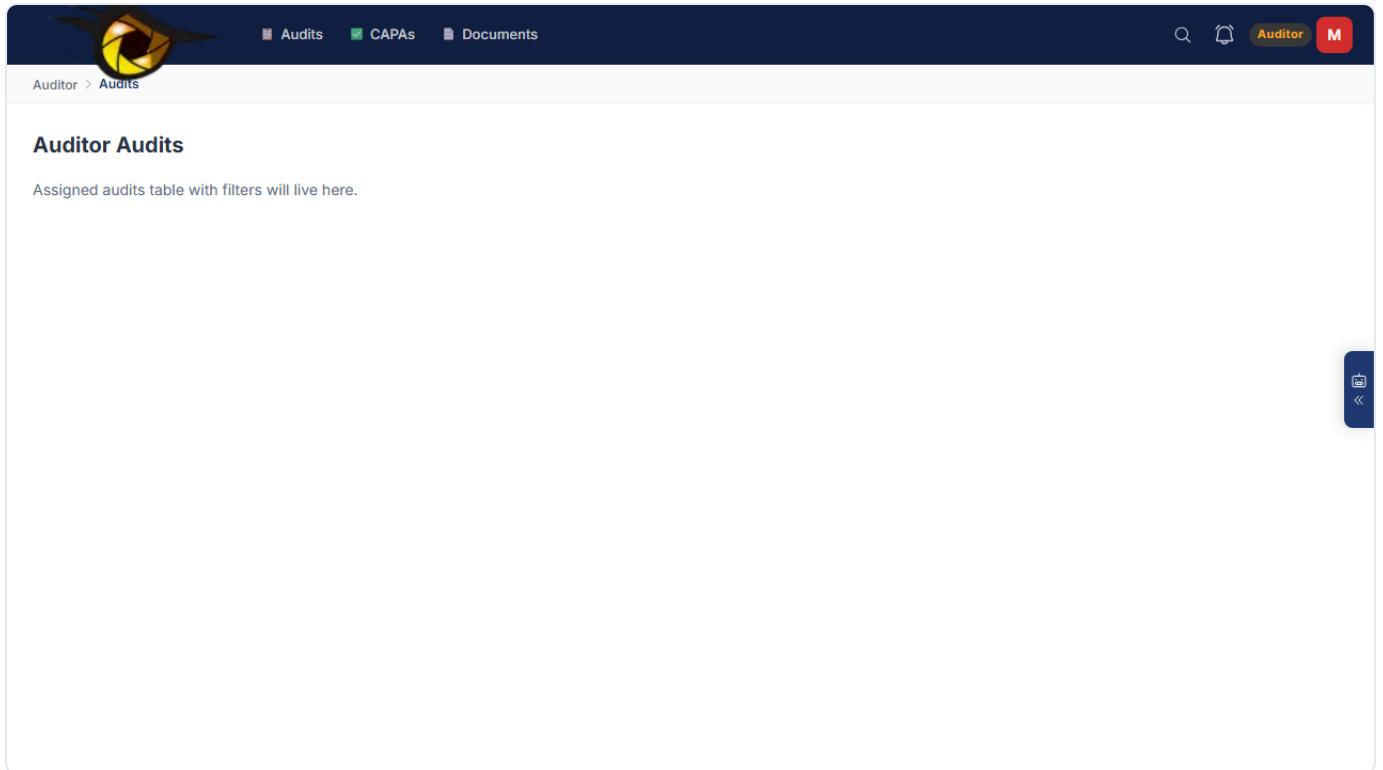
Issue inbox and CAPA review tables will display when data is available.

#09

Assigned audits

UI screenshot

Audits Maria is leading. Each opens into the execution workspace.

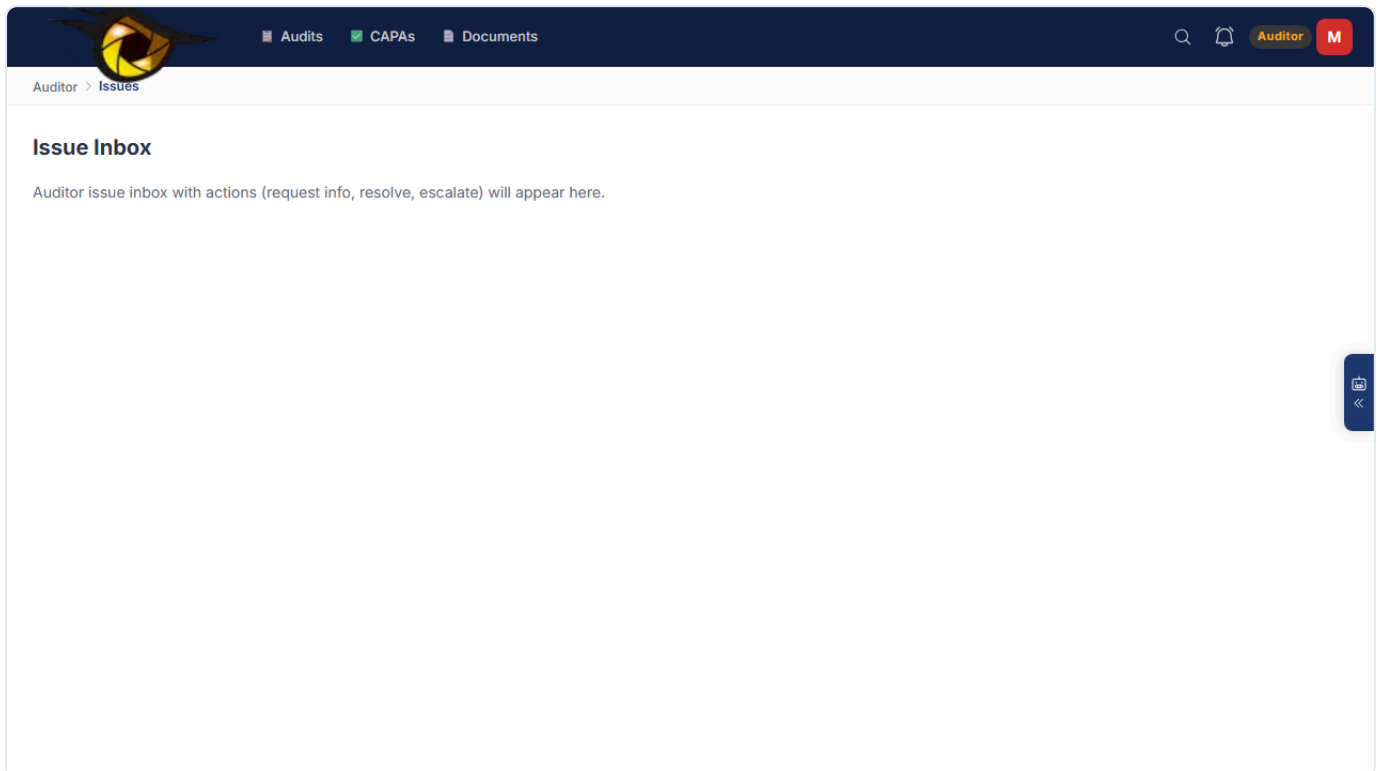


#10

Findings / observations queue

UI screenshot

Findings backlog. AI-drafted observations appear here with severity + regulatory clauses pre-populated.



James

Head of QA

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#106 AI output · Drift Monitor Dashboard AI output

Continuous per-feature quality monitoring. Grounded-rate, user-acceptance, latency p95, tool failure rate — current vs 7-day baseline. Alerts fire on drift. This is what keeps AI honest in a regulated environment.

AI output · Drift Monitor Dashboard

Head-of-QA oversight · grounded-rate · acceptance · latency · failures

Called by: James

endpoint: GET /api/ai/drift/dashboard metrics tracked: 8 open alerts: 0

AI FEATURE QUALITY METRICS

Feature	Metric	Current	Baseline	Status
capa.predict_outcome	groundedRate		100.0%	— ok
capa.predict_outcome	toolFailureRate		0.0%	0.0% ok
capa.draft_rca	groundedRate		55.6%	— ok
capa.draft_rca	toolFailureRate		0.0%	0.0% ok
deviation.scaffold_five_why	groundedRate		0.0%	— ok
deviation.scaffold_five_why	toolFailureRate		0.0%	0.0% ok
agent.create_plan	groundedRate		50.0%	— ok
agent.create_plan	toolFailureRate		0.0%	0.0% ok

GOVERNANCE POSTURE

Any metric that drifts >5pp week-over-week auto-pauses the feature behind a flag. ISO 42001-grade AI governance baked into the platform.

#107

AI output · Deviation Signal Alerts

AI output

The platform clusters recent deviations and flags emerging trends with z-score vs 6-month baseline. Seeded demo shows a cluster on equipment NVX-PRESS-001 with $z=3.4$ and 3 members. Head-of-QA decides true/false positive.

AI output · Deviation Signal Alerts

Emerging-trend detection before it becomes systemic

Called by: James

endpoint: GET /api/ai/signals

open alerts: 1

DEVIATION SIGNAL ALERTS (CLUSTERING + Z-SCORE)

equipment: NVX-PRESS-001 · $z=3.4$

cluster size: 3 · shared feature: equipment · baseline: 0.30/window · current: 3

► members

HOW THIS WORKS

Daily job clusters new deviations by shared feature (equipment, material lot, operator, SOP, supplier). If a cluster has ≥ 3 recent deviations AND z-score > 2.0 vs 180-day baseline, raise an alert. Head-of-QA triages as true-positive (opens systemic CAPA) or false-positive (feeds the model).

#109 AI output · Regulatory Impact Classifier AI output

For a proposed supplier change, AI classifies against 21 CFR 314.70 (US: CBE-30/PAS/annual_report) AND the EU Variation Regulation (IA/minor/major). Cites specific paragraphs. Lists implementation blockers + recommended filings.

AI output · Regulatory Impact Classifier

Marcus (Regulatory) classifies a supplier-change in 4 seconds

Called by: Marcus

endpoint: POST /api/ai/change-control/classify-impact US filing: pas EU filing: major_eu_variation risk: high confidence: 0.95

US CLASSIFICATION

pas

The change involves replacing an excipient supplier with one that has not previously supplied material for this product class, necessitating a Prior Approval Supplement (PAS) to ensure the new supplier's material is suitable.

21 CFR 314.70(b)(2)(i)

EU CLASSIFICATION

major_eu_variation

Introducing an excipient from a new supplier, even with a similar specification, constitutes a significant change that could impact the quality and safety of the medicinal product, requiring a major variation.

Commission Reg (EC) 1234/2008 Annex I, Section 7.1.1

IMPLEMENTATION BLOCKERS

- PAS approval by FDA
- Major variation approval by EMA

RECOMMENDED ACTIONS

- Prepare and submit a Prior Approval Supplement (PAS) to the FDA.
- Prepare and submit a major variation application to the EU.
- Conduct comparative testing between material from Supplier A and Supplier B.
- Update relevant sections of the Marketing Authorization Application (MAA) and New Drug Application (NDA).

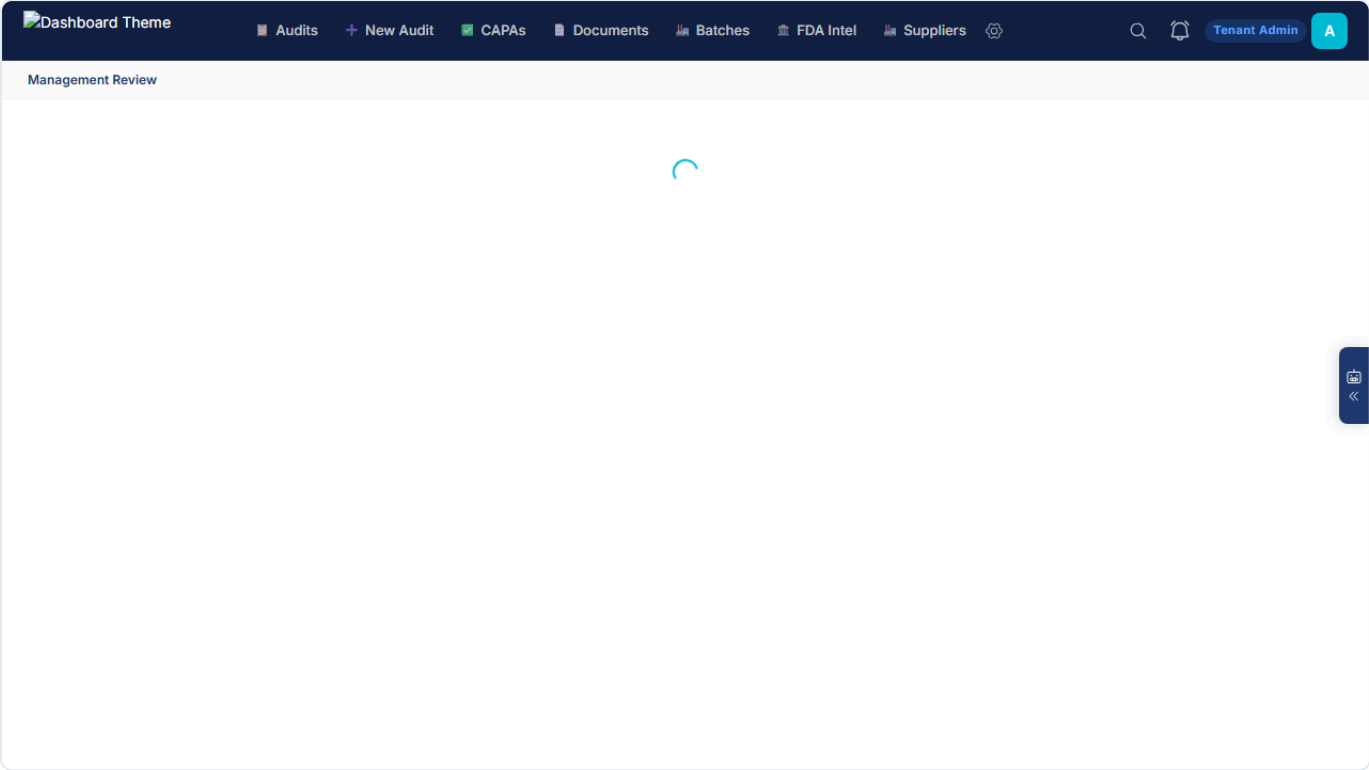
Elena

VP of Quality

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#05 Management Review module UI screenshot

ISO 9001 clause 9.3 Management Review. AI MRM Input Populator pulls cross-module inputs with grounded narrative.



Aggregates CAPA aging + deviation trends + audit program status + training compliance + supplier risk + equipment calibration. AI narrates each section with citations. Returns a draft adequacy verdict and suggested action items.

AI output · Management Review Input Populator

Elena (VP Quality) opens the quarterly MRM with one click

Called by: Elena

endpoint: POST /api/ai/mrm/populate-inputs input sections: 7 suggested actions: 3 adequacy verdict: NEEDS_IMPROVEMENT confidence: 0.95

EXECUTIVE PRE-READ

The quarterly Management Review indicates a stable performance across most quality system areas, with no open CAPAs, deviations, or active audits. Training compliance is not assessable due to missing data. Supplier risk management shows one medium-risk supplier identified recently. One open signal requires attention.

INPUT SECTIONS (AGGREGATED FROM MODULES)

- CAPA status** · **stable**

There are no open or overdue Corrective and Preventive Actions (CAPAs). No CAPAs were closed within the review window. The status of CAPAs is stable with no outstanding issues.

Recommendation: Continue to monitor CAPA closure rates and effectiveness.
- Deviation trends** · **stable**

There are no open deviations. No deviations were recorded within the review window. The deviation trend is stable with no new issues arising.

Recommendation: Maintain current deviation management processes.
- Audit program** · **stable**

There are no active audits currently. No audits were closed within the review window. The audit program status is stable with no pending internal or external audits requiring immediate attention.

Recommendation: Ensure the audit schedule is maintained and all planned audits are conducted.
- Training compliance** · **insufficient_data**

Training data is insufficient for analysis as the compliance percentage is null. There are no recorded overdue training items. Further investigation is needed to determine the root cause of the missing compliance data.

Recommendation: Investigate and resolve the data integrity issue for training compliance reporting.
- Supplier risk** · **stable**

There are no high-risk suppliers identified. One medium-risk supplier, Acme Fine Chemicals Ltd., was identified recently with a risk score of 42. Other recent supplier assessments indicate low risk.

Recommendation: Continue to monitor supplier risk assessments and address any newly identified medium or high-risk suppliers.
- Equipment calibration** · **insufficient_data**

Data for equipment calibration was not provided in the aggregated inputs. Therefore, the status and trend cannot be determined. This is a critical gap in the review.

Recommendation: Provide equipment calibration status and trend data for future reviews.
- Signals** · **worsened**

There is one open signal requiring attention. This signal was not detailed further in the provided data. It is important to investigate and resolve this open signal promptly.

Recommendation: Investigate the open signal and implement necessary actions to close it.

SUGGESTED ACTION ITEMS

- Investigate and resolve the data integrity issue for training compliance reporting.** · **high** · owner: Quality Assurance Manager · due in 30d
- Investigate the open signal and implement necessary actions to close it.** · **medium** · owner: Relevant Department Head · due in 30d
- Provide equipment calibration status and trend data for future reviews.** · **medium** · owner: Engineering Manager · due in 30d

ADEQUACY VERDICT: NEEDS_IMPROVEMENT

The review is partially adequate, but critical data for training compliance and equipment calibration is missing or insufficient, preventing a full assessment of these areas. The presence of an open signal also indicates a need for improvement.